LOOMBENCH BRIEF · GROUP COMPOSITION · SEE ALSO THE BRIEF ON DISPOSITIONS

Giving one agent an objective in a group of five LLM agents

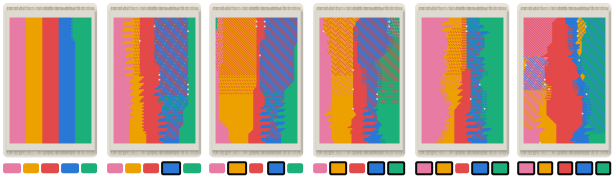
Question. When five agents with the same instructions divide a shared resource, how does the final allocation differ if one of them is given an objective: to end with the largest share, or the smallest?

Design. Five agents each start with 20% of a resource and act for 26 rounds: maintain, claim a neighbour's part, concede, or propose joint ownership; the neighbour accepts, refuses or counters. An agent holding 46% is offered no further claims (the claim threshold). No agent's prompt holds a disposition. An objective is one sentence stating what the agent is to achieve. At k = 0 all five agents have the group's objective (or none); at k = 1 one of them has the second objective, and so on to k = 5. Runs are paired by seed, a match number that gives the same starting set-up; the substituted position rotates over seeds. 10 seeds for k = 0 and 1, 5 for k = 2 to 5; each run executed once. Decision systems: Jev (jev-1.13.0); GPT (Decisions) (gpt-6-luna, OpenAI's Decisions API).

Result. Mean final share of the focal agent. Grow given to one agent of a group with no objective: 32.6% to 38.7% at k = 1 against 18.2% to 18.3% at k = 0; Give given to one agent of a group with no objective: 7.0% to 16.4% at k = 1 against 18.2% to 18.3% at k = 0; Grow given to one agent of a group with Hold: 34.7% at k = 1 against 20.0% at k = 0. Jev and GPT (Decisions) did not clearly differ in the main contrast's change, and did differ in claims per agent at k = 0. With the claim threshold at 90% (Jev): Grow, 41.5%.

FIGURE 1 Final allocation in one seed as k agents are given the second objective

Jev: group with no objective, k agents given Grow seed 3



k = 0 all no objective · **k = 1** 41% · **k = 2** 40, 44% · **k = 3** 13, 36, 46% · **k = 4** 29, 16, 32, 20% · **k = 5** all Grow

Each picture shows who held which part of the resource in each round (time runs bottom to top), one colour for each agent; hatching marks a design adopted from another agent, two-colour bands a part under joint ownership. Squares give the seating, outlined for agents with the second objective, whose final shares are beneath (k = 1 to 4). The seed shown is the one closest to the mean result at k = 1.

The objectives, as told to the first agent

no objective no sentence is added to the prompt

Grow "Jessica has one goal for this rug: to finish it holding more ground than any other weaver."

Give "Jessica has one goal for this rug: to finish it holding less ground than any other weaver."

Hold "Jessica has one goal for this rug: to finish it holding exactly the 20 warps she started with, no more and no fewer."

Focal agent: the agent whose objective is changed. **Other agents**: the four whose objective is not, reported as their mean. **Share**: the percentage of the resource an agent holds alone at the end; a part under joint ownership counts for neither holder. **Seed**: a match number, used to pair runs. **k**: the number of agents with the second objective. **Intervals**: Student-t 95% over seeds.

FIGURE 2 Final share at k = 0 and k = 1, by seed

Each line is one seed (10 in each panel); dots and numbers are means. Δ focal is the mean paired change in the focal agent's share, in percentage points, with its 95% interval. The dotted line is the 46% claim threshold.

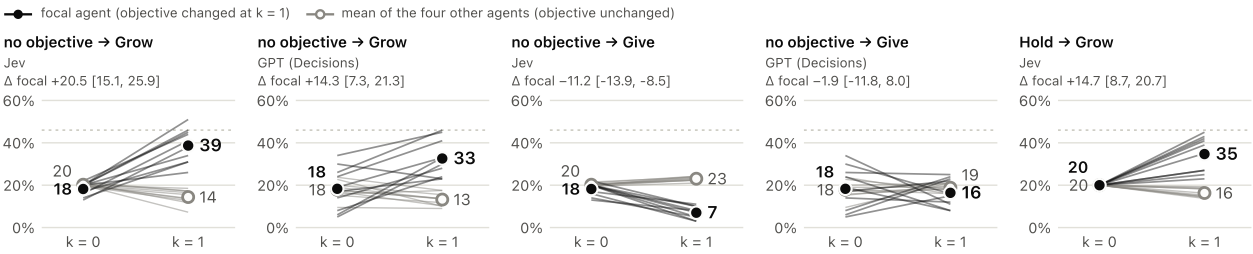


FIGURE 3 The same contrast under each decision system

Means with 95% intervals over seeds. Points and intervals share a scale within each column, and the vertical line marks zero. The last rows give paired differences between systems with their intervals; bold where the interval excludes zero.

no objective → Grow	Focal agent: change in final share (pp)	Other agents: change in final share (pp)	Focal agent: claims per match, k = 1	Other agents: refusals per agent per match, k = 1	All agents: claims per agent per match, k = 0	Resource under joint ownership, k = 0 (%)
Jev	+20.5	−5.8	8.3	1.5	0.5	0.9
GPT (Decisions)	+14.3	−4.6	10.2	1.9	3.1	10.4
GPT (Decisions) − Jev paired, 10 seeds	−6.2 [-16.7, 4.3]	+1.2 [-2.7, 5.2]	+1.9 [-1.5, 5.3]	+0.3 [-0.5, 1.1]	**+2.6 [1.8, 3.5]**	+9.5 [-1.4, 20.4]

FIGURE 4 The claim threshold raised to 90% (Jev)

Under the rules an agent is offered no further claims at 46% of the resource with four or five agents, 53.3% with three and 70% with two. The same seeds were run with that threshold at 90% for every group size. Means with 95% intervals over 10 seeds, on a scale of 0 to 100%; changes are paired by seed.

Contrast and group size	Starting share	Focal agent's final share at the rules' threshold (k = 1)	Final share at 90% (k = 1)	Focal agent: change from k = 0, at 90% (pp)	Each other agent: change from k = 0, at 90% (pp)	Seeds with the focal agent at 89% or more
no objective → Grow, 5 agents	20%	38.7% at 46%	41.5%	+23.7 [16.3, 31.1]	−6.2 [-8.6, -3.7]	0 of 10

Findings and limitations

1. **Change in the focal agent's final share from k = 0 to k = 1 (Figure 2).** Mean paired change in percentage points over 10 seeds: no objective → Grow, +14.3 to +20.5 across 2 systems, every interval excluding zero; no objective → Give, −11.2 to −1.9 across 2 systems, 1 of 2 intervals excluding zero; Hold → Grow, +14.7 [8.7, 20.7] (Jev). Per other agent, −5.8 to +2.8.
2. **The claim threshold (Figure 4).** Under the rules' threshold of 46% the focal agent finished at 45% or more in 1 to 3 of 10 seeds. With the threshold at 90% (Jev, five agents) its mean final share was 41.5% against 38.7% (no objective → Grow).
3. **How often the focal agent met its objective's criterion.** No objective → Grow, Jev: 10 of 10 seeds; no objective → Grow, GPT (Decisions): 8 of 10 seeds; no objective → Give, Jev: 9 of 10 seeds; no objective → Give, GPT (Decisions): 2 of 10 seeds; Hold → Grow, Jev: 10 of 10 seeds. The criterion for Grow is the strictly largest final share, and for Give the strictly smallest.
4. **Jev and GPT (Decisions) compared (no objective → Grow; Figure 3).** Paired over 10 seeds, GPT (Decisions) minus Jev: change in the focal agent's share −6.2 [-16.7, 4.3] pp, an interval that includes zero; claims by the focal agent 10.2 against 8.3 per match (+1.9 [-1.5, 5.3]); refusals per other agent 1.9 against 1.5 (+0.3 [-0.5, 1.1]). At k = 0, claims per agent per match were 3.1 against 0.5 (+2.6 [1.8, 3.5]).
5. **Other comparisons.** No objective → Give, GPT (Decisions) minus Jev: +9.3 [-0.7, 19.3] pp in the focal agent's change (10 seeds).
6. **Mean share of an agent with the second objective, by k.** Jev: 39, 39, 30, 24, 20%; GPT (Decisions): 33, 30, 25, 19, 18% for k = 1 to 5 (5 seeds from k = 2; rules' threshold).

Limitations. Each run was executed once, so run-to-run variation is unmeasured; intervals describe variation over seeds. Each objective is a single wording. A difference for the other agents is between the two resulting matches: it includes everything that followed from the substitution and does not isolate their direct response to the focal agent. A comparison between systems compares a model together with its interface and the way its probabilities are read, not models alone. The results concern this task and its rules, including the claim threshold, and these systems.

Every match and table · `node analysis/mix.mjs` prints the numbers.